

An Investigation of Hallucination in LLM-Based Recommender Systems

DS535 Course Project

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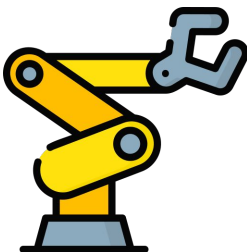
Findings

Future Directions

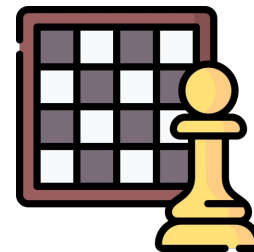
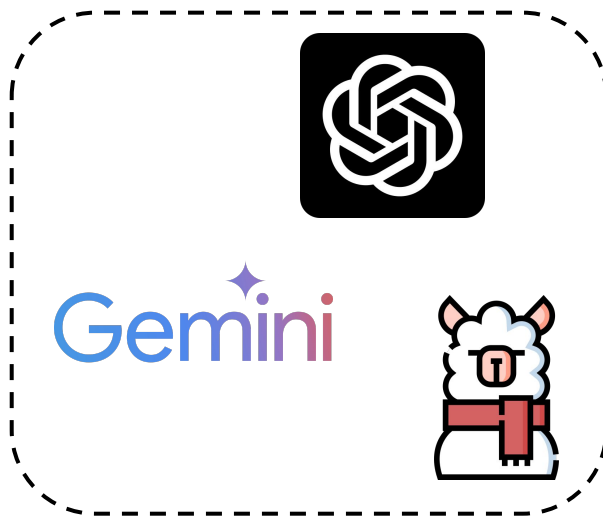
Language Models for Decision Making



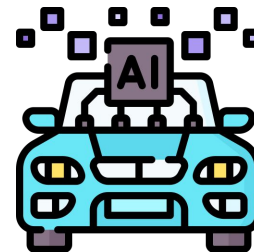
web browsing



robotic manipulation



game playing



autonomous driving

Language Models for Recommendation

Sequential Recommendation raw data for *Beauty*

```
user_id: 7641      user_name: Victor
purchase_history: 652 -> 460 -> 447 -> 653 -> 654 -> 655 -> 656 -> 8
-> 657
next_item: 552
candidate_items: 4885 , 4280 , 4886 , 1907 , 870 , 4281 , 4222 ,
4887 , 2892 , 4888 , 2879 , 3147 , 2195 , 3148 , 3179 , 1951 ,
..... , 1982 , 552 , 2754 , 2481 , 1916 , 2822 , 1325
```

Here is the purchase history of user_{{user_id}}:
{{purchase_history}}
What to recommend next for the user?

{{next_item}}

(b)

Direct Recommendation raw data for *Beauty*

```
user_id: 250      user_name: moriah rose
target_item: 520
random_negative_item: 9711
candidate_items: 4915 , 1823 , 3112 , 3821 , 3773 , 520 , 7384 ,
7469 , 9318 , 3876 , 1143 , 789 , 595 , 3824 , 3587 , 10396 ,
..... , 2766 , 7498 , 2490 , 3232 , 9711 , 2975 , 1405 , 8051
```

Choose the best item from the candidates to recommend for
{{user_name}}? \n {{candidate_items}}

{{target_item}}

(c)

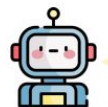
A Fundamental Problem: **Hallucination!**

User Input



Can you recommend a delicious recipe for dinner?

LLM Response



Yes, here is a delicious recipe for **lunch**. So how about fried chicken with mashed potatoes? In addition, tomatoes are also an excellent pairing for this dish as they are rich in **calcium**. Enjoy this **steak**!

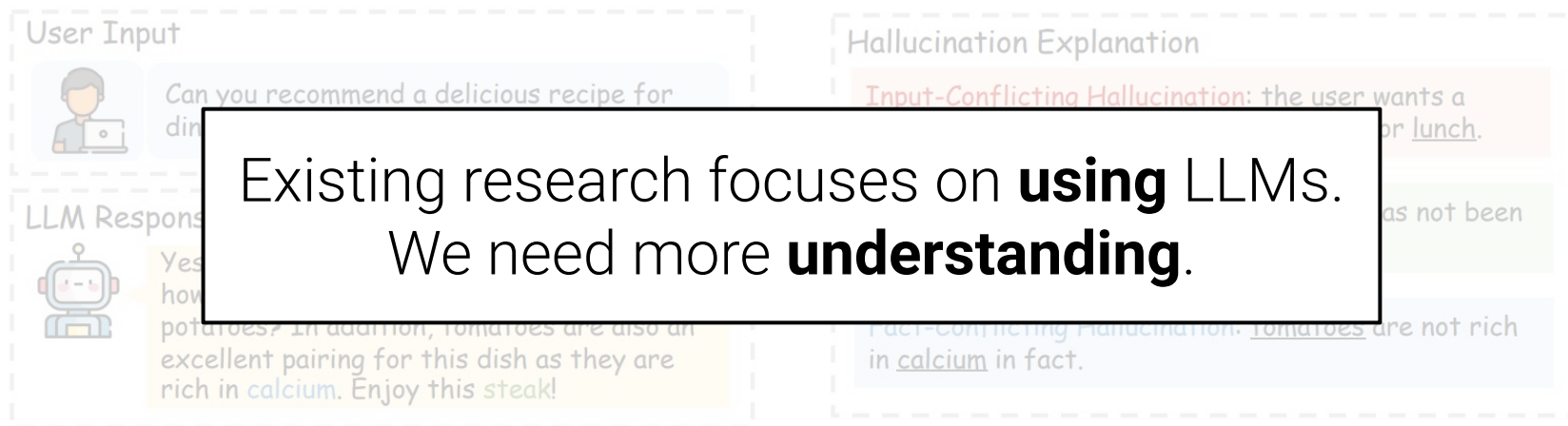
Hallucination Explanation

Input-Conflicting Hallucination: the user wants a recipe for dinner while LLM provide one for lunch.

Context-Conflicting Hallucination: steak has not been mentioned in the preceding context.

Fact-Conflicting Hallucination: tomatoes are not rich in calcium in fact.

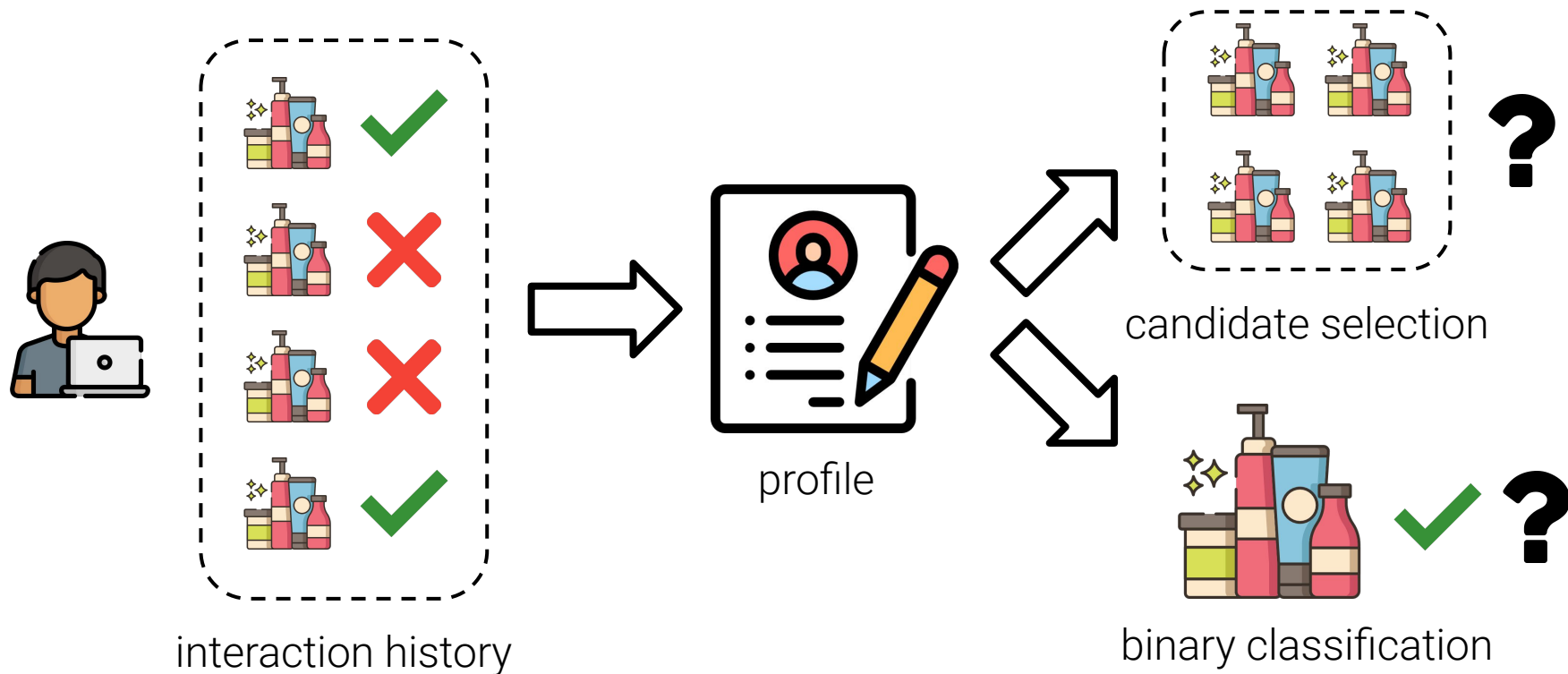
A Fundamental Problem: **Hallucination!**



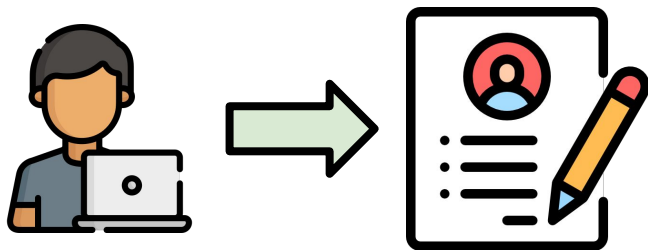
Existing research focuses on **using** LLMs.
We need more **understanding**.

What is **hallucination** in the context
of **recommender systems**?

Our Formulation: Sequential Recommendation



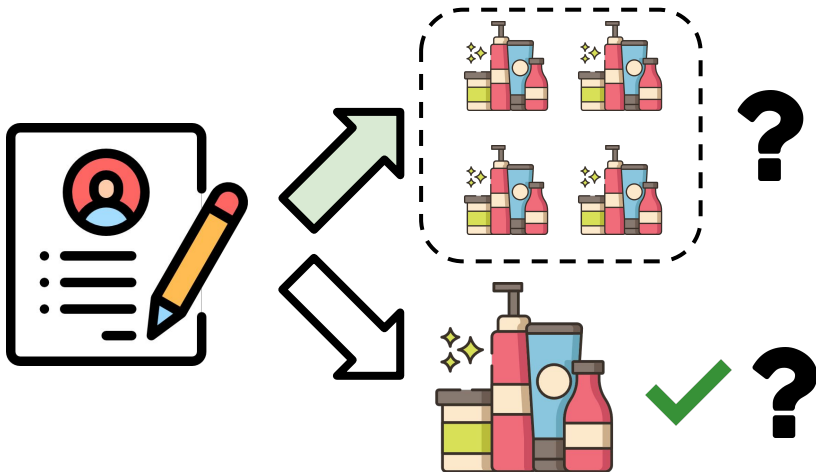
Types of Hallucination



Type 1

misalignment between profile & history

e.g., profile contains information not relevant to history



Type 2

misalignment between recommended products & profile

e.g., product does not exist or is not supported by profile

Research Questions

RQ1: How does type 1 hallucination affect recommendation performance?

RQ2: How does type 2 hallucination affect recommendation performance?

RQ3: What is the relation between type 1 and type 2 hallucination?

Experimental Setup: Experiments

E0: Compare between different profile generation schemes

RQ1: How does type 1 hallucination affect recommendation performance?

E1: Measure correlation between type 1 hallucination and recommendation performance

RQ2: How does type 2 hallucination affect recommendation performance?

E2: Measure correlation between type 2 hallucination and recommendation performance

RQ3: What is the relation between type 1 and type 2 hallucination?

E3: Measure correlation between type 1 and type 2 hallucination

Experimental Setup: Data & Hallucination Metrics

What data to use? 

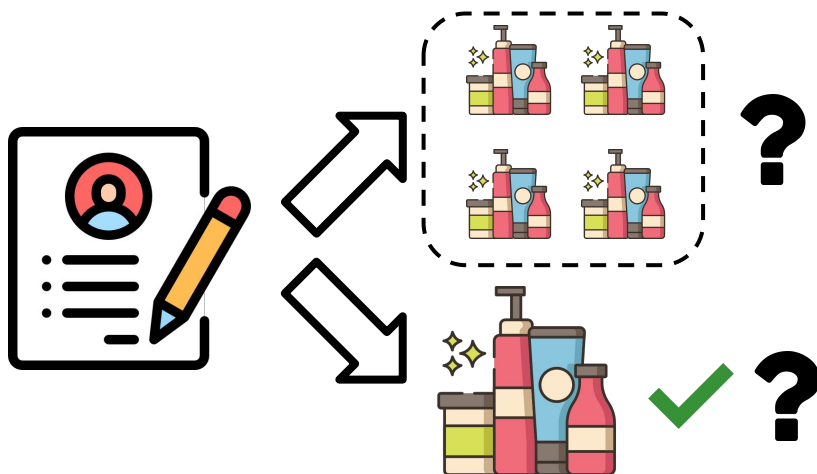
“All Beauty” & “Movies and TV”
500 users for each category
Each with ≥ 5 and ≤ 20 interactions

Which hallucination metrics to use?



Experimental Setup: Recommender Metrics

Which recommender metrics to use?

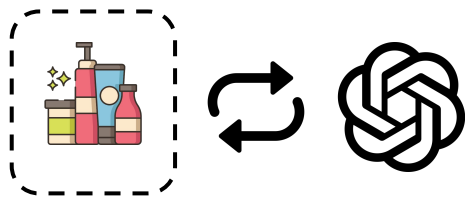


NDCG@1, NDCG@5, NDCG@10
Hit Rate @1, Hit Rate @5, Hit Rate @10

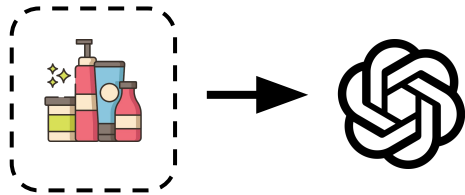
True Positive Rate
True Negative Rate
Overall Accuracy

Experimental Setup: Profile Generation

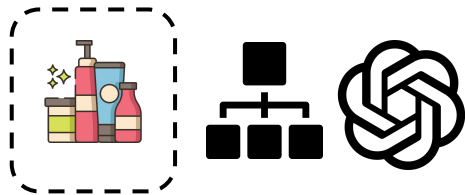
How are profiles generated?



Iterative: LLM loops through products one-by-one



Vanilla: history is fed directly to LLM



Vanilla Structured: Vanilla but with a structure requirement

What language model to use?



GPT-4o-mini

Experimental Setup: Experiments

E0: Compare between different profile generation schemes

RQ1: How does type 1 hallucination affect recommendation performance?

E1: Measure correlation between type 1 hallucination and recommendation performance

RQ2: How does type 2 hallucination affect recommendation performance?

E2: Measure correlation between type 2 hallucination and recommendation performance

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Experiments

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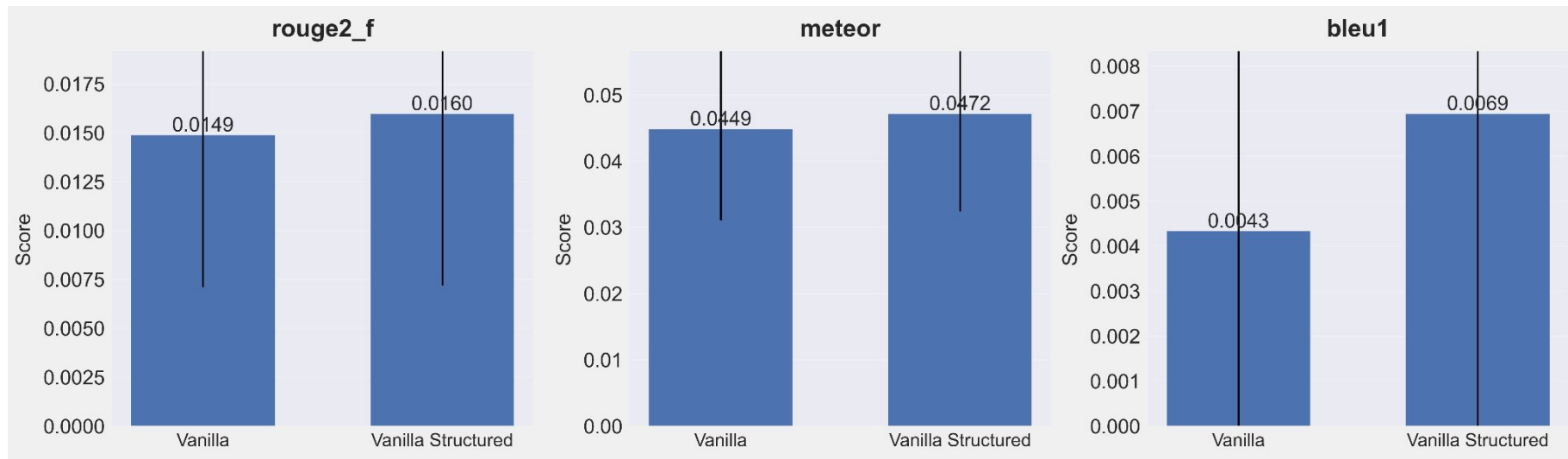
E1: Measure correlation between type 1 hallucination and recommendation performance

E2: Measure correlation between type 2 hallucination and recommendation performance

E3: Measure correlation between type 1 and type 2 hallucination

Results - Hallucination in Profile Generation

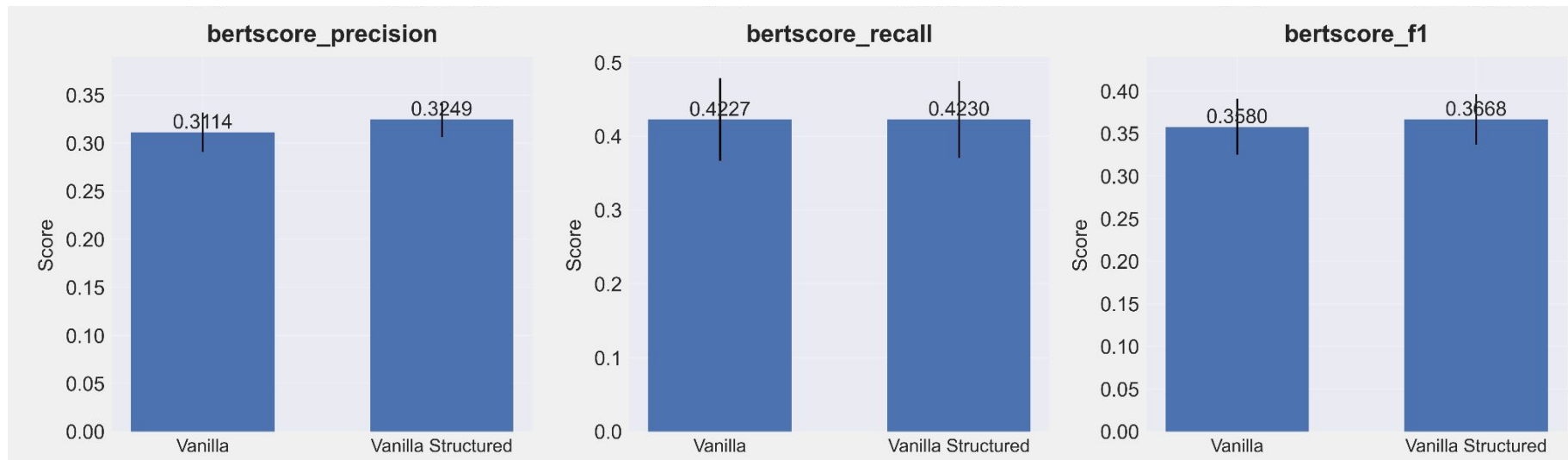
Beauty Dataset



LLM's ability to retain **lexical similarity** is enhanced as the prompt becomes structured

Results - Hallucination in Profile Generation

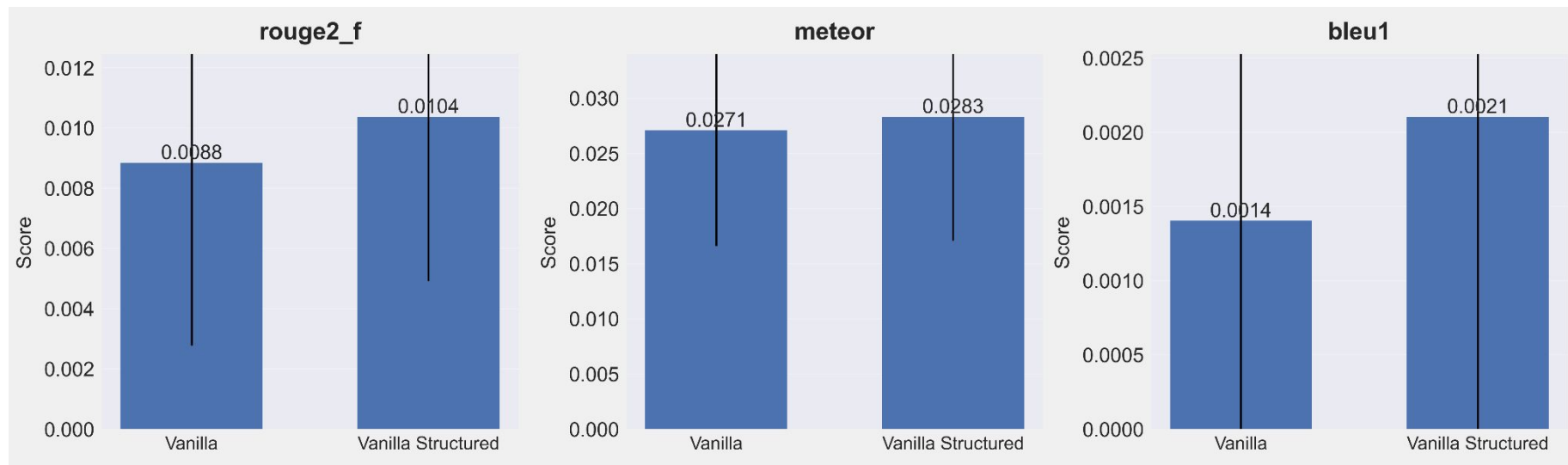
Beauty Dataset



Minute improvements in **semantic retainment** as prompt becomes structured

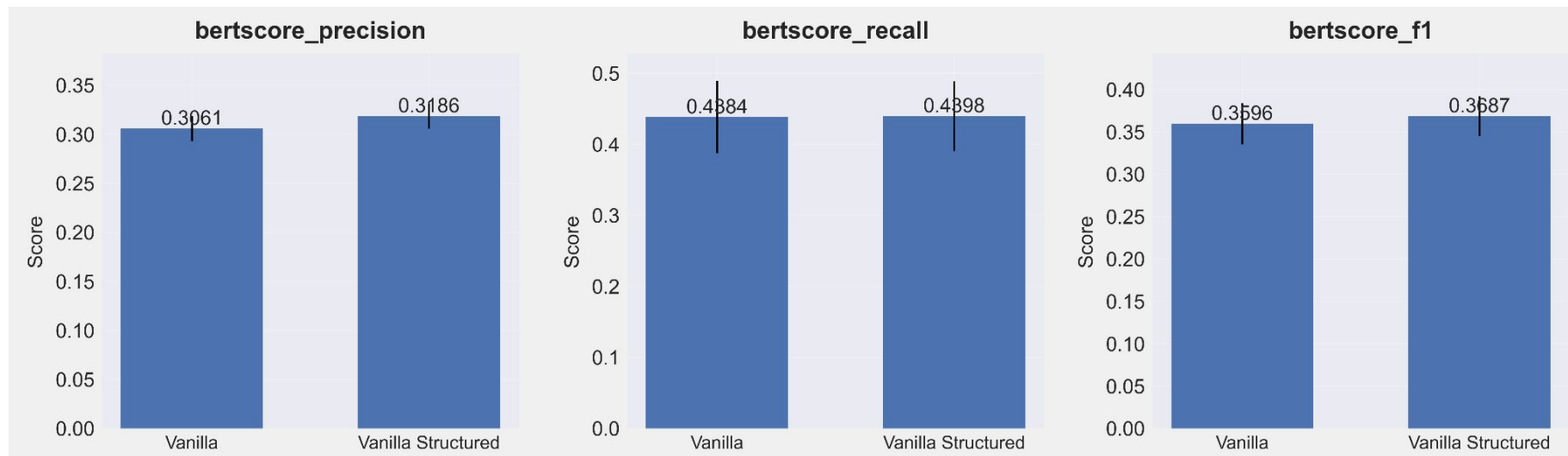
Results - Hallucination in Profile Generation

Movies Dataset



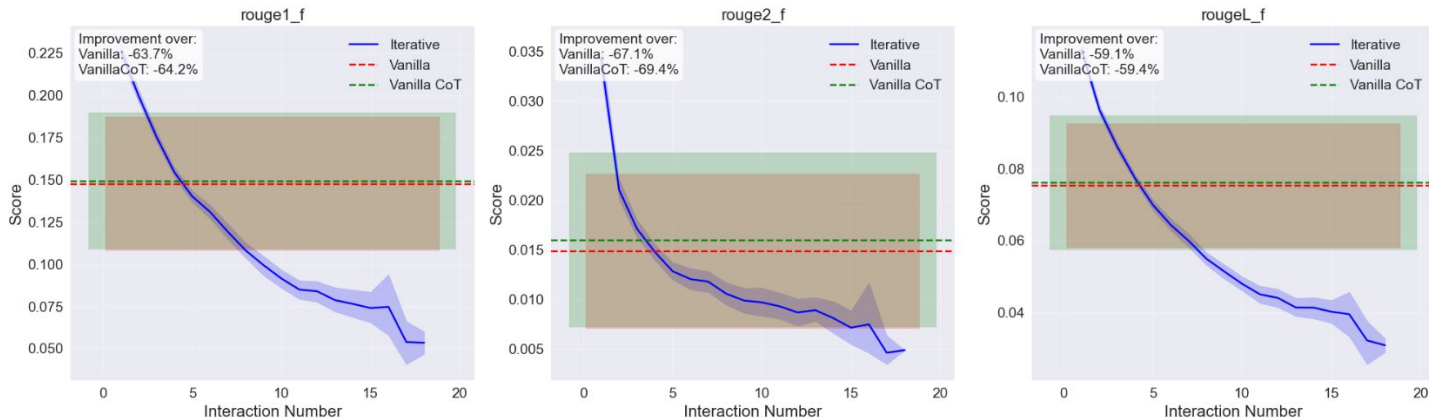
Results - Hallucination in Profile Generation

Movies Dataset

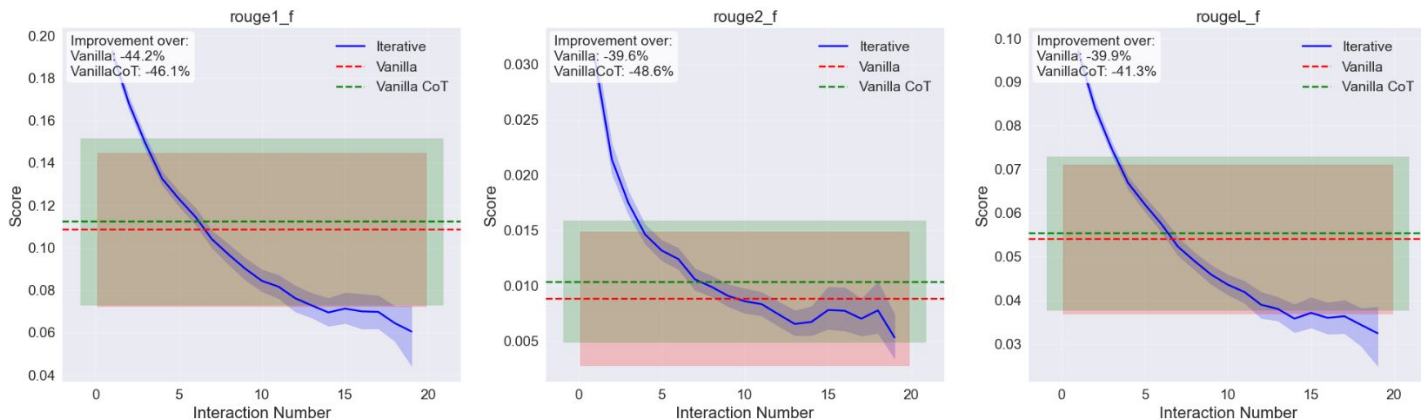


Results - Hallucination in Profile Generation

Beauty Dataset

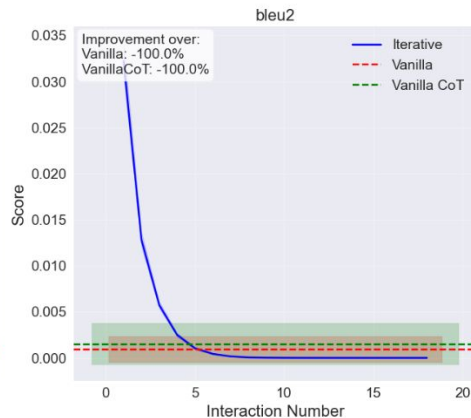
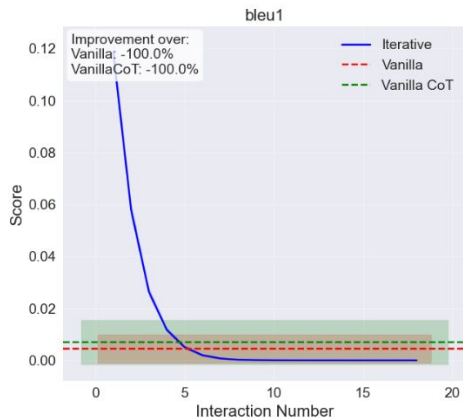
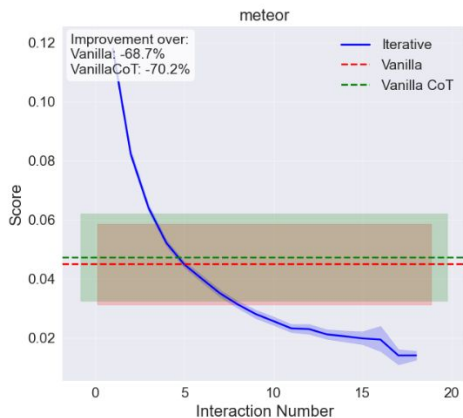


Movie Dataset

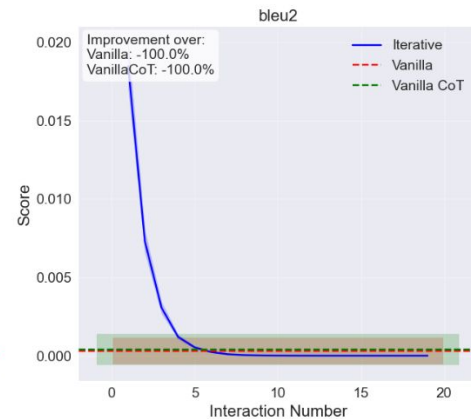
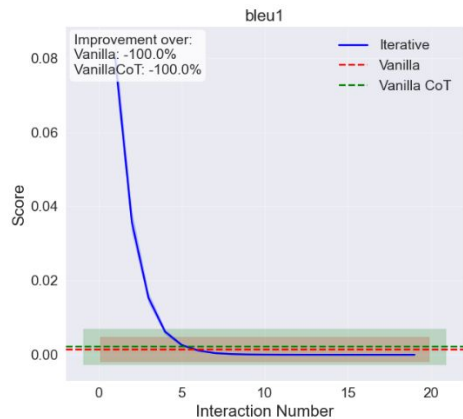
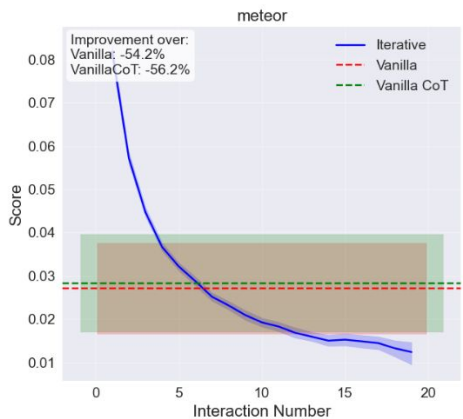


Results - Hallucination in Profile Generation

Beauty Dataset

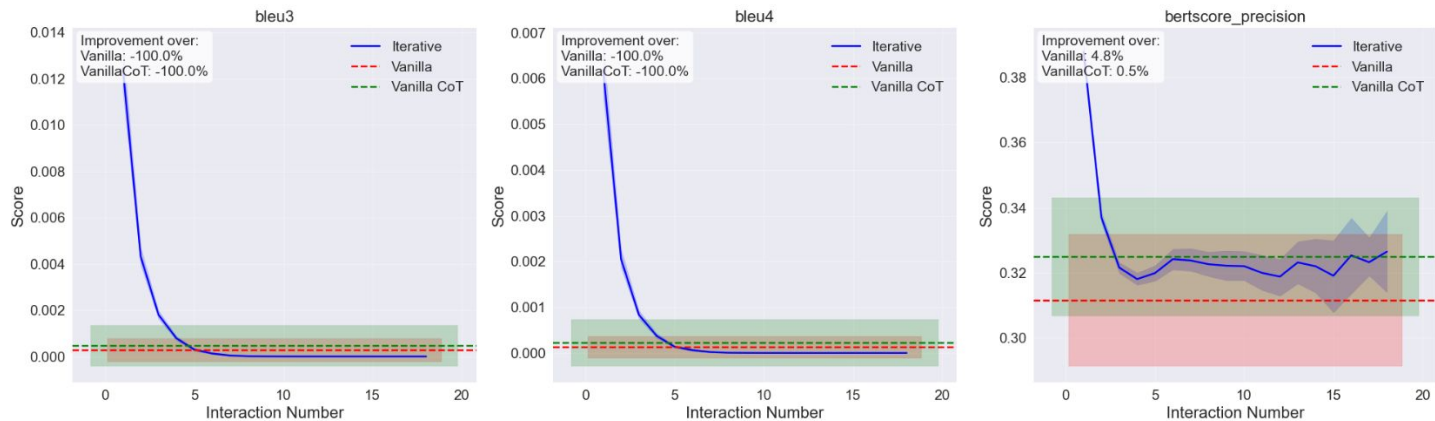


Movie Dataset

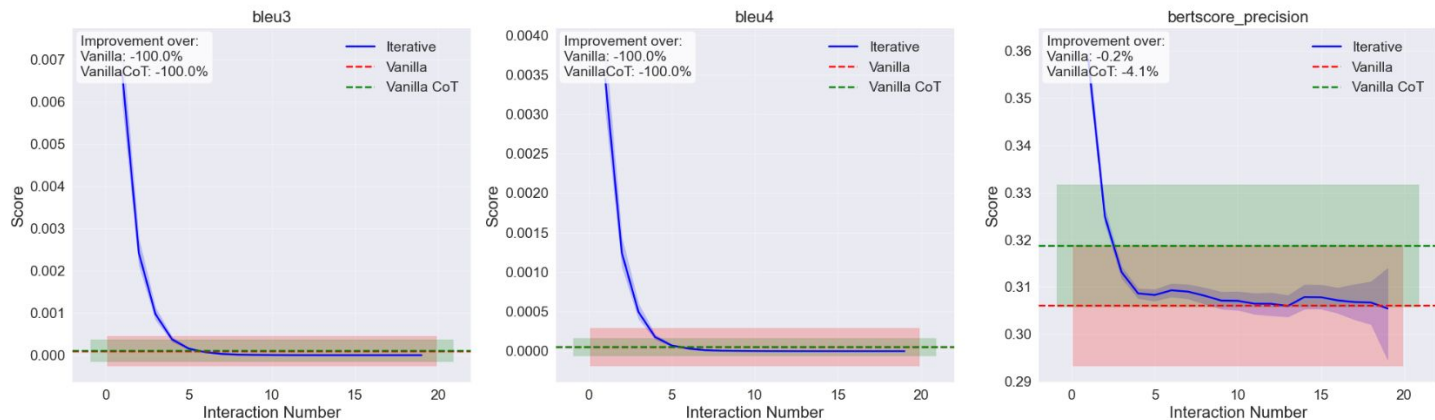


Results - Hallucination in Profile Generation

Beauty Dataset

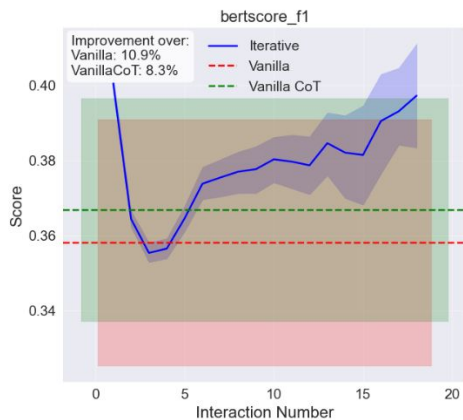
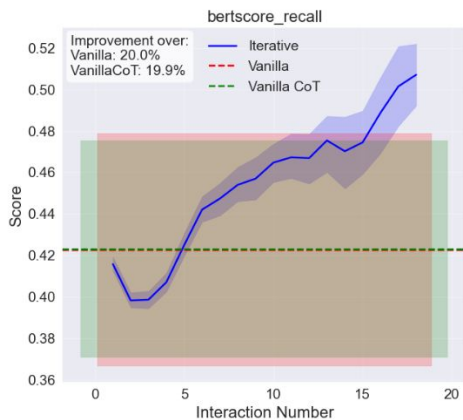


Movie Dataset

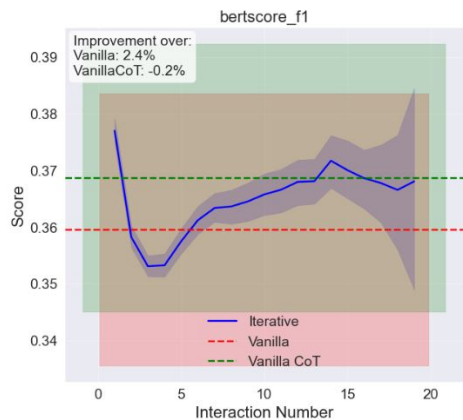
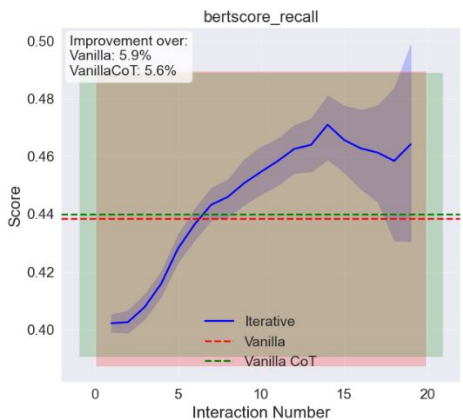


Results - Hallucination in Profile Generation

Beauty Dataset



Movie Dataset



**Enhanced Semantic
Retainment ability**

Experiments

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RQ3: What is the relation between type 1 and type 2 hallucination?

E0: Compare between different profile generation schemes

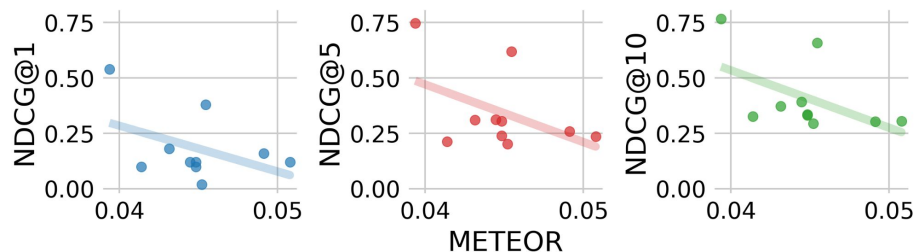
E1: Measure correlation between type 1 hallucination and recommendation performance

E2: Measure correlation between type 2 hallucination and recommendation performance

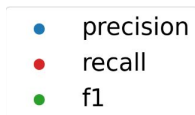
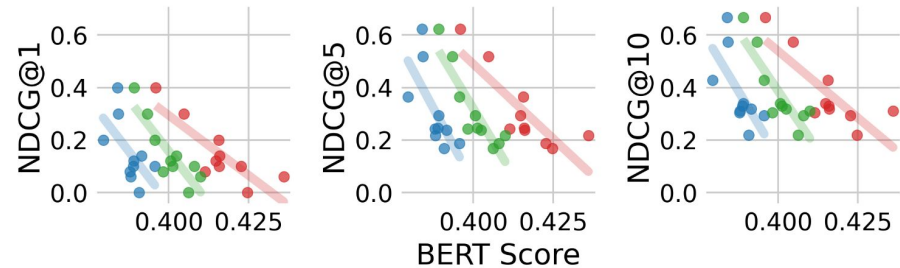
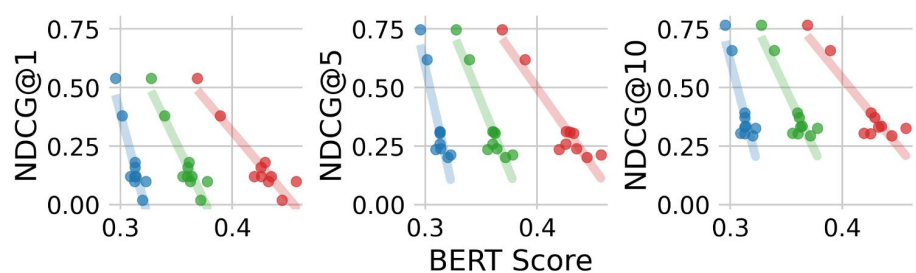
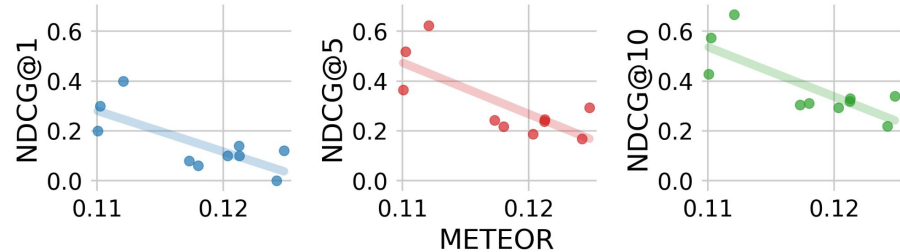
E3: Measure correlation between type 1 and type 2 hallucination

Results - Decrease in Performance as Type 1 Hallucination Decreases

Vanilla Profile on Beauty Dataset

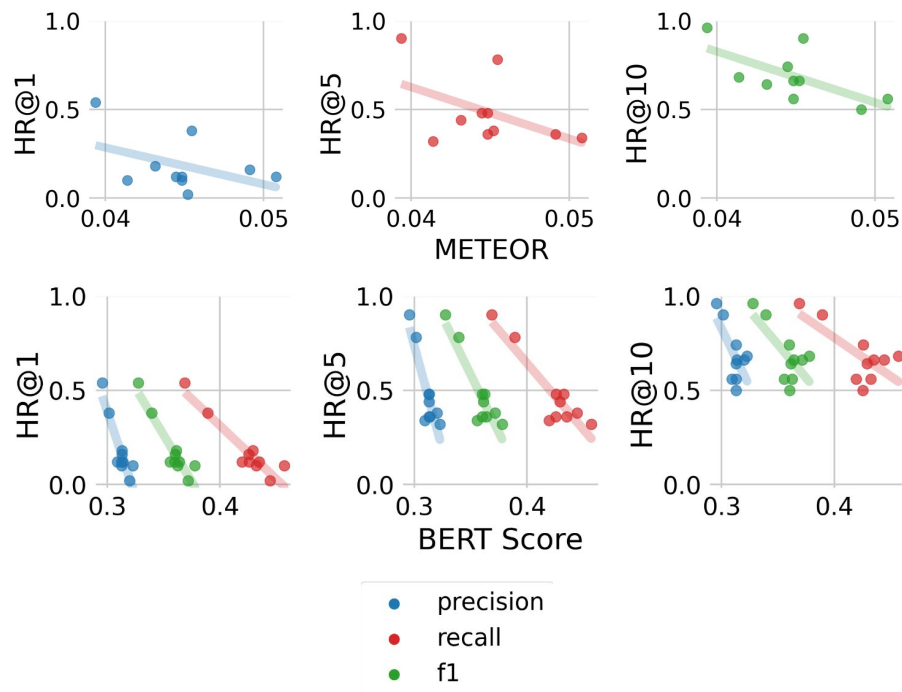


Iterative Profile on Beauty Dataset

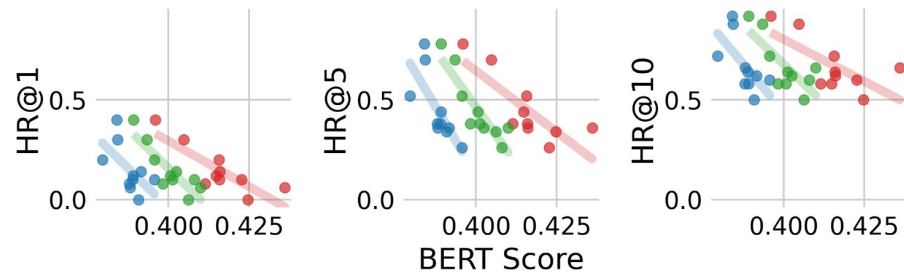


Results - Decrease in Performance as Type 1 Hallucination Decreases

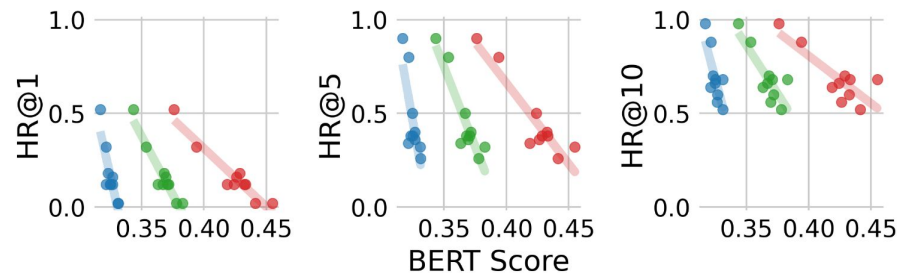
Vanilla Profile on Beauty Dataset



Iterative Profile on Beauty Dataset

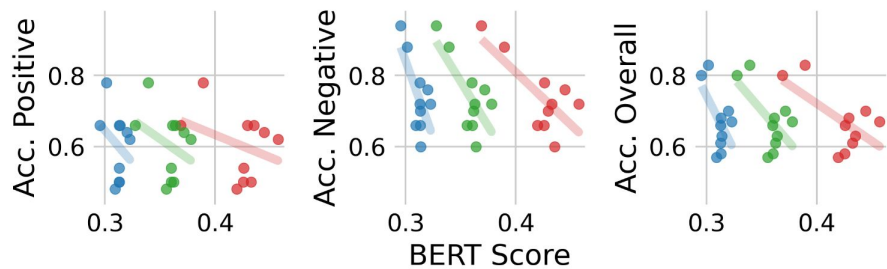


Vanilla Structured Profile on Beauty Dataset

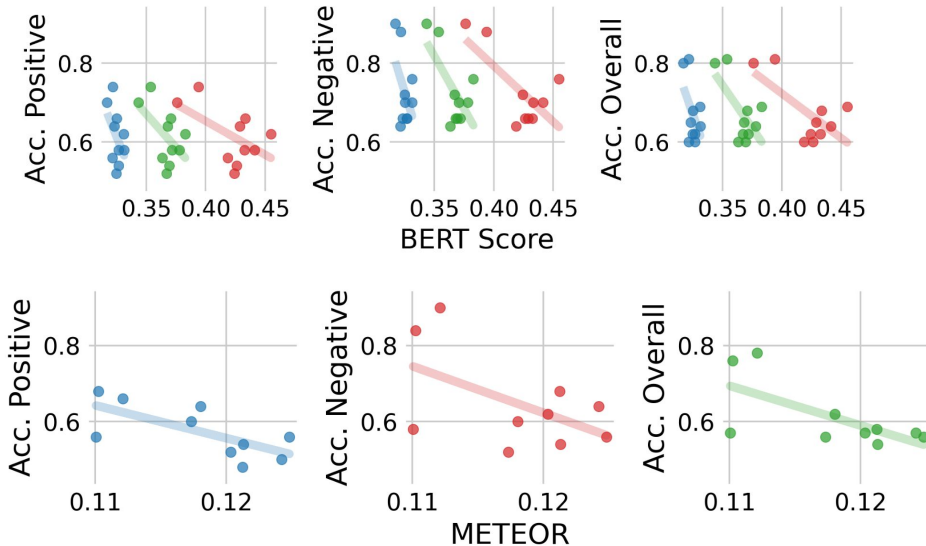


Results - Decrease in Performance as Type 1 Hallucination Decreases

Vanilla Profile on Beauty Dataset



Vanilla Structured Profile on Beauty Dataset



Experiments

RQ1: How does type 1 hallucination affect recommendation performance?

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RQ3: What is the relation between type 1 and type 2 hallucination?

E0: Compare between different profile generation schemes

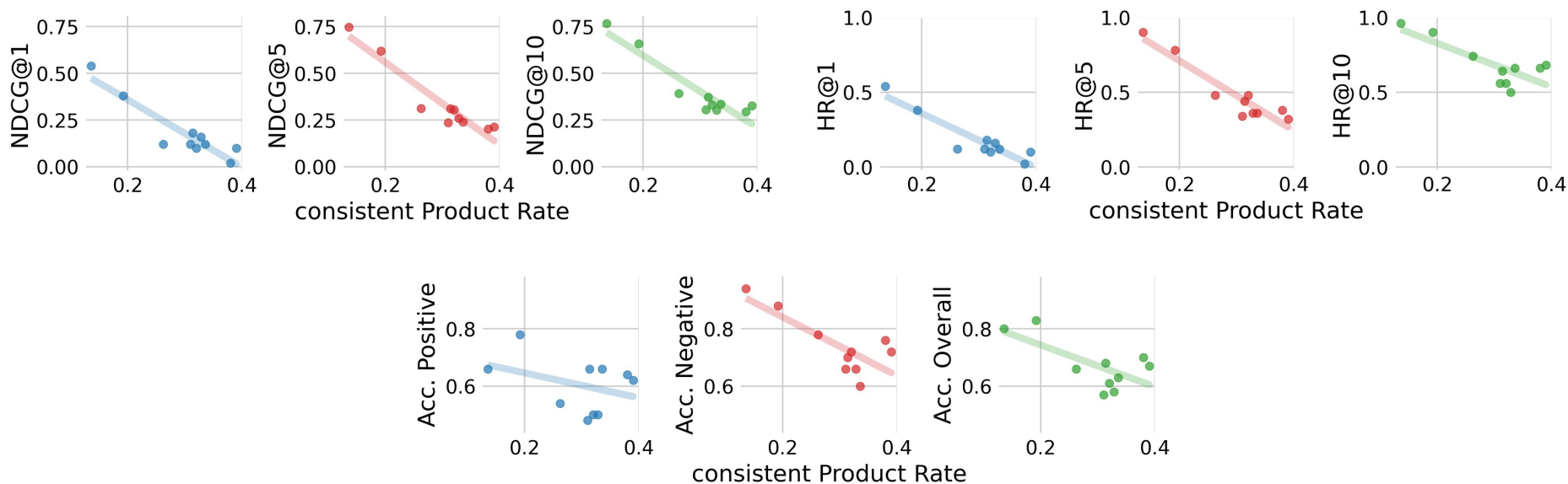
E1: Measure correlation between type 1 hallucination and recommendation performance

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E3: Measure correlation between type 1 and type 2 hallucination

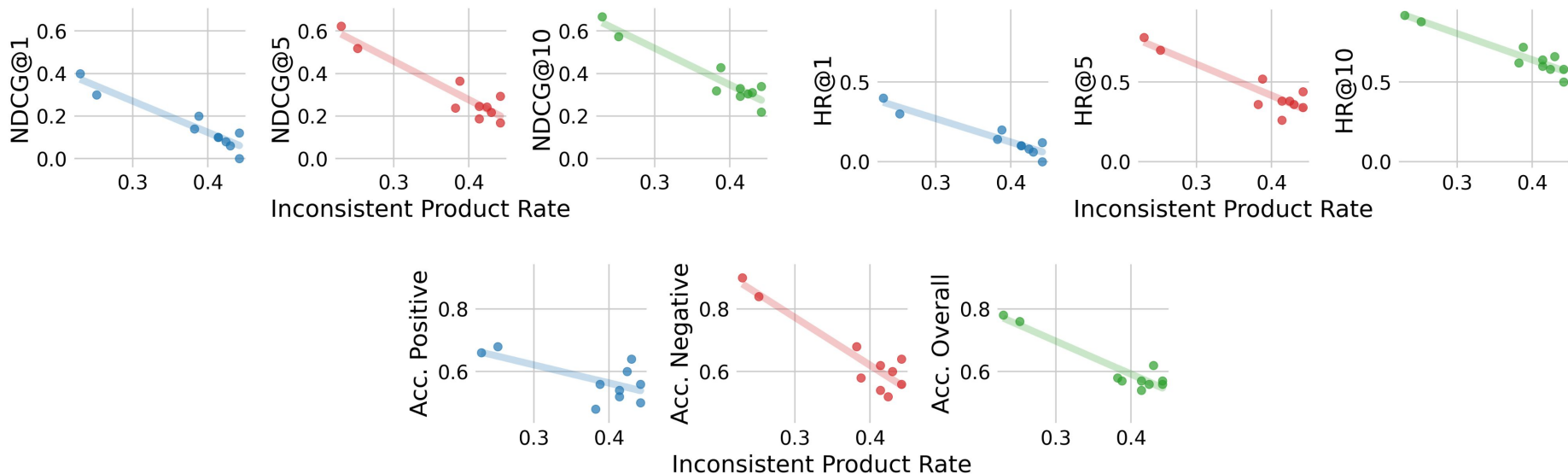
Results - Decrease in Performance as Type 2 Hallucination Decreases

Performance on Vanilla Profile for Beauty Dataset



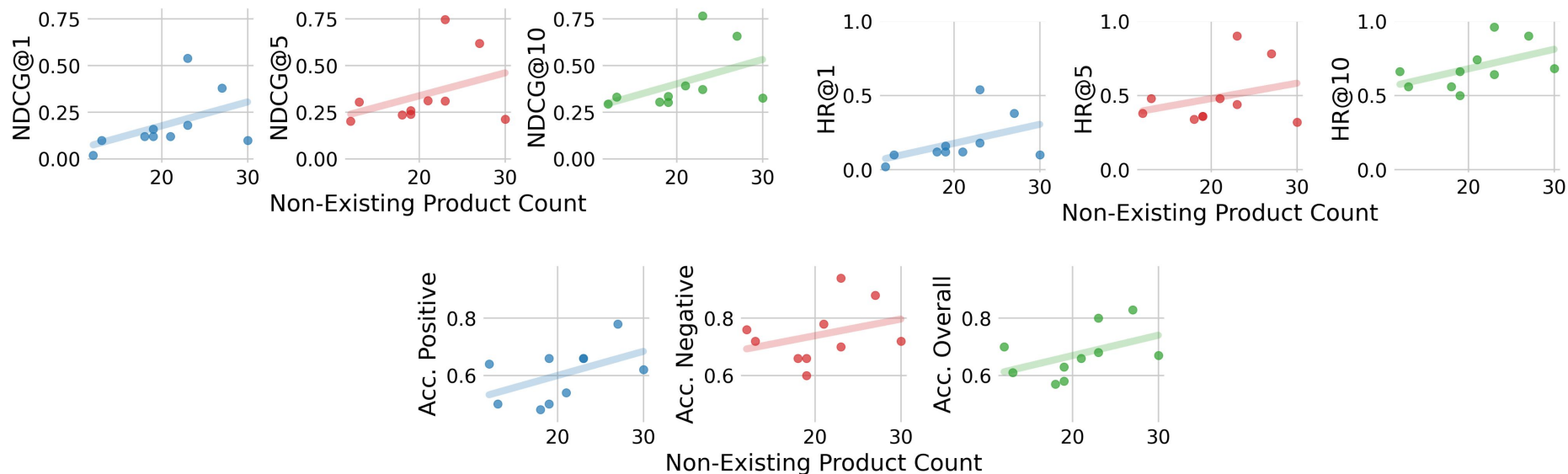
Results - Decrease in Performance as Type 2 Hallucination Decreases

Performance on Iterative Profile for Beauty Dataset



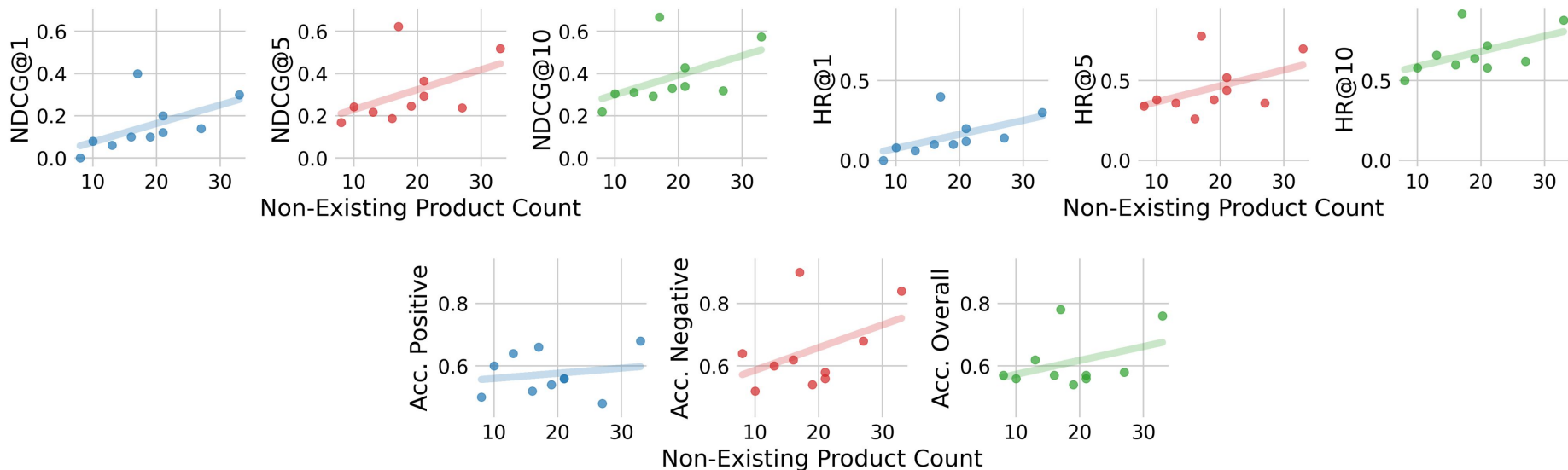
Results - Increase in performance as more “NE” products exist

Performance on Vanilla Profile for Beauty Dataset



Results - Increase in performance as more “NE” products exist

Performance on Iterative Profile for Beauty Dataset



Experiments

RQ1: How does type 1 hallucination affect recommendation performance?

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RQ3: What is the relation between type 1 and type 2 hallucination?

E0: Compare between different profile generation schemes

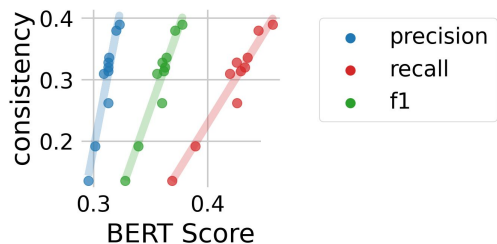
E1: Measure correlation between type 1 hallucination and recommendation performance

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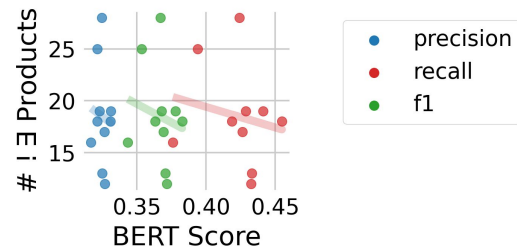
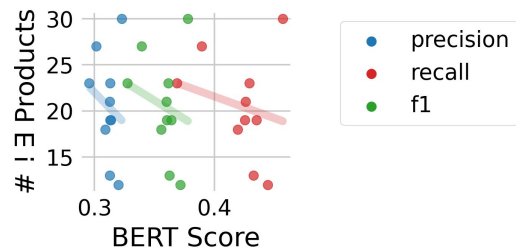
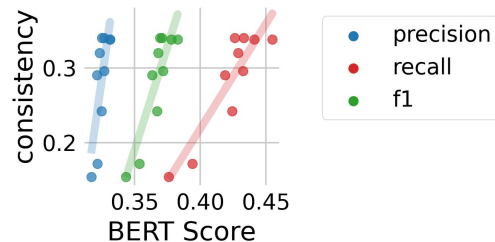
E3: Measure correlation between type 1 and type 2 hallucination

Results - No Correlation Except BERT Score and Consistency

Performance of Vanilla Profile for Beauty Dataset



Performance of Vanilla Structured Profile for Beauty Dataset



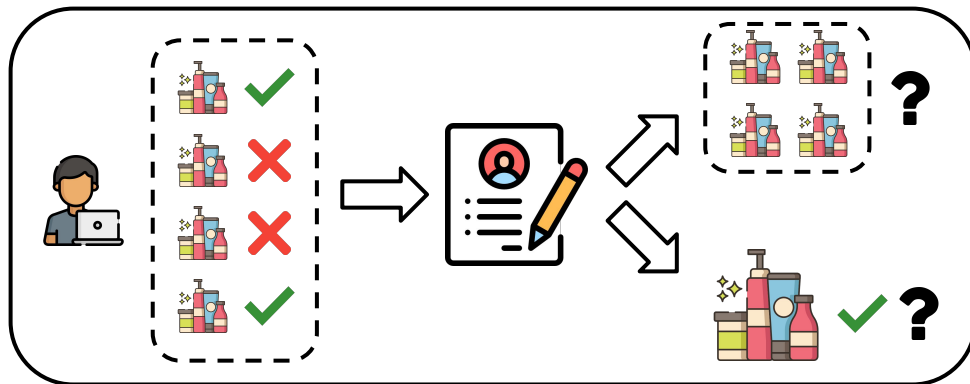
Takeaway

1. Hallucination in profile generation is **negatively correlated** with recommendation performance
2. Hallucination in recommendation generation is (surprisingly) **positively correlated** with recommendation performance
3. Hallucination in profile generation mostly has **no correlation** with hallucination in recommendation

Where to go from here?

1. From recommender to hallucination

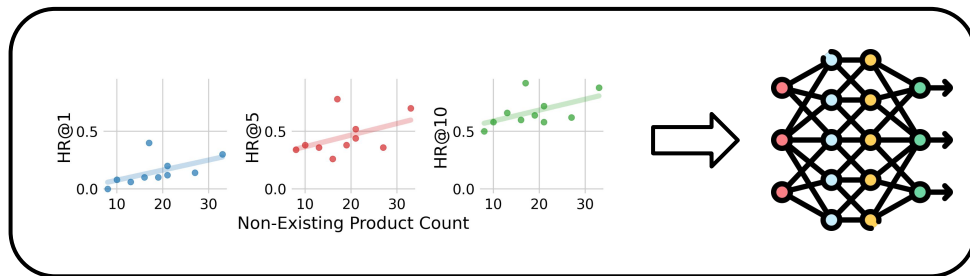
How do model choice and training mechanism affect hallucination and the impact of hallucination?



This is not the only LLM-based recommender!

2. From hallucination to recommender

How can we leverage hallucination information to design effective recommendation models?



What to do with this information?